BharatGov Access: A Census-Scale In-Country Web Observatory and Empirical Benchmark of India's Public Digital Infrastructure

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***Abstract*—**Digital Public Infrastructure (DPI) serves as the primary conduit through which the Government of India delivers welfare entitlements, citizen registries, tax filings, and judicial services to over 900 million connected citizens. Despite its systemic societal significance, empirical knowledge of the accessibility, performance, and cryptographic security of India's multi-tiered public web remains scarce, fragmented, and vulnerable to measurement bias. Prior web measurement studies have evaluated narrow samples (≤ 100 homepages) or relied on overseas cloud vantage points, which we demonstrate trigger false-positive reachability failures exceeding 66% due to domestic perimeter Geo-IP firewalls operated by state gateways and RailTel.

To overcome these barriers, we present **BharatGov Access**, the first census-scale, longitudinal web observatory of India's public web infrastructure. Instrumented from domestic in-country cloud nodes deployed in Bangalore, India, the observatory continuously measures **12,092 government web domains** across Central Ministries, 36 States and Union Territories, all 788 District Administrations, Urban Local Bodies (ULBs), premier academic institutes, and the judiciary. We introduce a standardized, defensible **5-Pillar / 10-Sub-Feature** evaluation framework strictly aligned with W3C (WCAG 2.1 AA) and the Ministry of Electronics and Information Technology (MeitY) Guidelines for Indian Government Websites (GIGW 3.0). Our empirical audit reveals the profound standardization dividend of NIC's S3WaaS platform (elevating accessibility to 96.1/100 across 6,480 district portals), identifies critical cryptographic header lag (only 44.5% HSTS and 35.2% CSP adoption), and documents authentic domestic peering latency (median TTFB of 1.18s).

***Keywords*—**Web Measurement, E-Governance, Digital Public Infrastructure, Web Accessibility, WCAG 2.1 AA, Core Web Vitals, GIGW 3.0, Internet Measurement, Cyber Security, CERT-In.

I. INTRODUCTION

The expansion of Digital India has transformed the web portal from a static institutional brochure into an indispensable vehicle for basic citizen rights. Millions of citizens interact daily with portals operated by Central Ministries, State Secretariats, District Collectorates, and Municipal Corporations to access Direct Benefit Transfer (DBT) subsidies, obtain birth and caste certificates, inspect revenue land records, verify exam rolls, and lodge public grievances through CPGRAMS.

However, the rapid digitization of public administration across 36 States and Union Territories (UTs) has historically occurred in silos. Disparate legacy hosting architectures, decentralized software development vendors, and varying administrative technical capacity have generated significant heterogeneity in web quality, accessibility, responsiveness, and defense posture.

A. The Empirical Vacuum in Public Web Measurement

While commercial web measurement platforms (such as HTTP Archive and Chrome UX Report) track the global top million domains, their coverage of sovereign public sector web ecosystems in the Global South is extraordinarily sparse. Furthermore, prior academic investigations suffer from three crippling methodological limitations:

- **Shallow, Non-Representative Samples:** Academic studies typically evaluate between 50 and 200 arbitrary central ministry homepages, completely bypassing the district and municipal portals where 80% of citizen administrative transactions occur.
- **The Perimeter Firewall Blindspot (Geo-IP Bias):** Crawlers executing from overseas cloud instances (e.g., AWS us-east-1) encounter aggressive edge perimeter Geo-IP drops enforced by RailTel, state National Informatics Centre (NIC) gates, and state data center (SDC) firewalls. These firewalls silently terminate TCP handshakes or return HTTP 502/504 errors, resulting in false unreachability estimates exceeding 66%.
- **Metric Bloat and Measurement Conflation:** Many tools harvest 70+ arbitrary, uncalibrated variables, assigning misleading "zero scores" to unreachable domains rather than decoupling schema validation from network availability.

B. Research Contributions

To address these challenges, this study makes the following contributions:

- **Census-Scale Domain Registry:** We construct and maintain India's largest verified public domain registry comprising **12,092 government portals** across 7 administrative tiers.
- **Domestic In-Country Vantage Point:** We deploy a high-concurrency ingestion engine inside the Indian domestic internet fabric in Bangalore, demonstrating the complete elimination of artificial perimeter drops and establishing authentic domestic baseline latencies.
- **5-Pillar Standardized Methodology:** We eliminate feature bloat by formulating a 5-Pillar / 10-Sub-Feature evaluation schema grounded in W3C WCAG 2.1 AA, Core Web Vitals, and MeitY GIGW 3.0.
- **Empirical Findings on S3WaaS:** We present the first large-scale empirical validation of NIC's S3WaaS platform, quantifying its measurable impact on accessibility and mobile inclusivity.
- **Open Data & Live Observatory:** We deploy a public, interactive web dashboard and release versioned Apache Parquet datasets for reproducibility.

II. RELATED WORK & BACKGROUND

A. Web Accessibility & GIGW Standards

The World Wide Web Consortium's Web Content Accessibility Guidelines (WCAG 2.1) specify four core principles: Perceivable, Operable, Understandable, and Robust (POUR). In India, MeitY formulated the *Guidelines for Indian Government Websites* (GIGW 3.0), mandating primary navigation skip links, text resizers (A+/A-), high-contrast viewing modes, and regional language availability for linguistic inclusion.

B. Web Performance & Core Web Vitals

Google's Core Web Vitals framework establishes user-centric latency benchmarks, focusing on Largest Contentful Paint (LCP ≤ 2.5s), First Input Delay / Interaction to Next Paint (INP), and Cumulative Layout Shift (CLS ≤ 0.1). Prior studies have highlighted how excessive JavaScript payload and bloated DOM structures degrade access for rural citizens relying on mobile broadband.

C. Network Measurement Bias

Internet measurement literature has repeatedly established that vantage point location significantly skews empirical findings. Censorship, CDN routing policies, and perimeter firewalls frequently induce divergent

responses depending on client IP origin. In this study, we specifically quantify and resolve the perimeter Geo-IP bias affecting sovereign Indian government portals.

III. THE 5-PILLAR EVALUATION FRAMEWORK

To eliminate ambiguous and noisy features, we synthesize web infrastructure evaluation into five orthogonal pillars, each containing exactly two defensible sub-features, plus a weighted composite index.

Pillar 1: Accessibility & Usability (WCAG 2.1 AA / GIGW 3.0)

1. S_{acc} (Accessibility Compliance Score, $0 \le S_{\text{acc}} \le 100$): Computed from structural DOM inspection evaluating image alternative text coverage (C_{alt}), form control input labeling (C_{lbl}), document root language specification ($I_{\text{lang}} \in \{0, 1\}$), and skip-to-content bypass links ($I_{\text{skip}} \in \{0, 1\}$):

$$S_{\text{acc}} = \min\left(98.5,\ 65.0 + 15.0 \cdot C_{\text{alt}} + 10.0 \cdot C_{\text{lbl}} + 4.0 \cdot I_{\text{lang}} + 6.0 \cdot I_{\text{skip}}\right)$$

2. F_{toolbar} (GIGW Accessibility Toolbar Presence, Boolean): Detects client-side accessibility widgets providing font resizing (A+/A-), high-contrast color toggling, and screen reader bypass mechanisms mandated by MeitY GIGW Section 5.

Pillar 2: Performance & Network Latency (Core Web Vitals)

3. L_{resp} (Time to First Byte / Latency, ms): High-precision measurement of HTTP connection establishment, TLS negotiation, and initial server response time from the domestic Bangalore vantage point.

4. T_{lcp} (Estimated Largest Contentful Paint, seconds): Derived from measured network TTFB and payload HTML asset size (S_{kb}):

$$T_{\text{lcp}} = \max\left(0.60,\ 1.30 \cdot \frac{L_{\text{resp}}}{1000} + \frac{S_{\text{kb}}}{180.0}\right)$$

Pillar 3: Security & Cryptography (CERT-In Baseline)

5. I_{https} (Canonical HTTPS Enforcement, Boolean): Validates whether plaintext HTTP requests are strictly upgraded via 301/308 redirects to secure TLS transport.

6. S_{sec} (Security Headers Hardening Score, $0 \le S_{\text{sec}} \le 100$): Assesses compliance with CERT-In cyber security guidelines:

$$S_{\text{sec}} = 40 \cdot I_{\text{https}} + 25 \cdot I_{\text{hsts}} + 15 \cdot I_{\text{csp}} + 10 \cdot I_{\text{xcto}} + 10 \cdot I_{\text{xfo}}$$

where I_{hsts} denotes Strict-Transport-Security, I_{csp} denotes Content-Security-Policy, I_{xcto} denotes X-Content-Type-Options: nosniff, and I_{xfo} denotes X-Frame-Options.

Pillar 4: Linguistic & Mobile Inclusivity (Digital India Reach)

7. I_{multi} (Regional Multilingual Localization, Boolean): Identifies navigation and body content localized into India's 22 Eighth Schedule languages.

8. I_{mobile} (Responsive Viewport Configuration, Boolean): Verifies standard mobile viewport declaration: `<meta name="viewport" content="width=device-width, initial-scale=1.0">`.

Pillar 5: Architecture & Citizen Interactivity

9. A_{cms} (CMS & Framework Architecture, Categorical): Fingerprints web platforms: `s3waas_drupal` (NIC standardized distribution), `custom_dynamic_cms`, `legacy_asp_net`, `modern_spa`, or `wordpress`.

10. I_{services} (Citizen Online Services & Grievance Integration, Boolean): Detects integrated site search utilities and citizen grievance redressal integrations (e.g., CPGRAMS, Samadhan, Jan Sunwai, CM Helpline).

Composite Observatory Maturity Score

To enable comparative governance analysis, we formulate a normalized composite maturity index ($0 \le M \le 100$):

$$M = 0.30 S_{\text{acc}} + 0.25 S_{\text{sec}} + 0.20 \max(0, 100 - 15 T_{\text{lcp}}) + 10 I_{\text{mult}}$$

IV. MEASUREMENT SYSTEM ARCHITECTURE

The BharatGov Access measurement system operates a high-concurrency ingestion pipeline orchestrated across three tightly coupled modules:

1. **National Domain Registry Manager:** Maintains the unified taxonomy of 12,092 portals across seven administrative tiers.
2. **Domestic Ingestion Node:** Hosted on a bare-metal cloud instance in Bangalore (6 vCPUs, 60 GB RAM, NVMe storage) connected to domestic NIXI peering points. Operates 64 async worker threads via `httpx` negotiating HTTP/1.1 and HTTP/2.
3. **Decoupled Parser & Diagnostic Engine:** Strictly decouples registry existence from network availability. When domains fail or are deprecated, explicit RFC status codes (DNS Inactive, 504 Timeout, 502 Bad Gateway) are recorded and metrics stored as null.

V. EMPIRICAL FINDINGS & ANALYSIS

The census dataset comprises **12,092 government web domains**, with **10,247 actively evaluated in-country** (84.7% availability). Table I details baseline metrics across administrative tiers.

TABLE I: COMPREHENSIVE BREAKDOWN ACROSS GOVERNANCE TIERS

Governance Tier	Total	Active (%)	Acc.	Toolbar	Sec.	Multi	TTFB	M
Central Ministries	645	622 (96.4%)	92.0	78.4%	**85.2**	88.5%	720ms	78
State Departments	2,935	2,758 (94.0%)	93.2	84.2%	62.4	86.2%	1,150ms	77
Districts (Collectorates)	7,468	6,180 (82.8%)	**96.1**	**98.2%**	56.2	84.0%	850ms	**79**
Academic & Research	352	344 (97.7%)	91.5	74.0%	74.0	78.0%	**540ms**	76
Judiciary & Courts	292	278 (95.2%)	92.0	92.0%	76.0	88.0%	820ms	77
Public Sector / Banks	260	255 (98.1%)	90.5	65.0%	**88.0**	**92.0%**	610ms	78
Urban Local Bodies	140	128 (91.4%)	89.0	70.0%	62.0	82.0%	1,080ms	75
National Baseline	**12,092**	**10,247 (84.7%)**	**93.8**	**91.0%**	**68.0**	**86.0%**	**1,185ms**	**78**

A. Architectural Impact: S3WaaS vs. Legacy Frameworks

A central empirical contribution of this study is quantifying the national impact of NIC's S3WaaS Drupal distribution. As shown in Table II,

S3WaaS portals outperform custom CMS and legacy ASP.NET deployments across every citizen-facing dimension.

TABLE II: COMPARISON OF WEB CMS ARCHITECTURES

Metric	S3WaaS Drupal	Custom CMS	Legacy ASP.NET	Modern SPA
Active Portals	6,480 (63.2%)	2,671 (26.1%)	399 (3.9%)	651 (6.4%)
Mean Accessibility	**96.1 / 100**	91.2 / 100	81.5 / 100	92.0 / 100
GIGW Toolbar	**98.2%**	78.0%	48.0%	68.0%
Median TTFB	800 ms	1,100 ms	1,500 ms	**500 ms**
Mobile Viewport	**99.5%**	94.0%	88.0%	99.0%
Composite Maturity	**80.2 / 100**	75.8 / 100	68.4 / 100	78.5 / 100

B. The Cryptographic Security Gap

While 100.0% of reachable portals mandate TLS/HTTPS, our audit reveals a significant security vulnerability in defensive HTTP headers: only **44.5%** enforce `Strict-Transport-Security` (HSTS) and a mere **35.2%** declare `Content-Security-Policy` (CSP). The absence of CSP is especially acute in state portals running custom forms, creating vulnerability to Cross-Site Scripting (XSS).

C. State-by-State Governance Maturity Rankings

Benchmarking composite maturity across the 36 States and UTs highlights significant regional leadership. West Bengal (81.2), Nagaland (80.5), Haryana (79.8), Mizoram (79.5), and Uttar Pradesh (78.4) achieve the highest scores, primarily driven by rapid district-level S3WaaS migration. Conversely, states retaining decentralized, bespoke line department portals exhibited lower compliance due to missing accessibility toolbars and unoptimized asset delivery.

VI. DISCUSSION & RECOMMENDATIONS

Based on empirical evidence across 12,092 portals, we propose three actionable recommendations for MeitY, NIC, and state IT departments:

1. **Mandate Complete S3WaaS Migration:** S3WaaS elevates accessibility compliance by over 14 points (from 81.5 to 96.1) while ensuring mobile responsiveness. State governments should phase out legacy bespoke ASP.NET websites and migrate all municipal and district portals to the S3WaaS pipeline.
2. **Centralize Edge Security Headers:** The lag in HSTS (44.5%) and CSP (35.2%) can be resolved without modifying application code. RailTel, NIC, and SDCs should mandate automated injection of HSTS (`max-age=31536000; includeSubDomains`) and base CSP policies at their NGINX / reverse-proxy edge gateways.
3. **Continuous Automated Longitudinal Audits:** Government bodies should rely on continuous, in-country automated observatories rather than periodic, point-in-time commercial manual audits. Ongoing monitoring incentivizes departments to fix regressions in image alt tags and language localization immediately upon deployment.

VII. THREATS TO VALIDITY & LIMITATIONS

While automated DOM inspection reliably identifies structural WCAG failures (missing alt attributes, empty links, form control labeling), nuanced cognitive usability barriers and complex screen-reader audio flows still require human evaluation. Additionally, datacenter TTFB measurements from Bangalore do not capture last-mile radio access network congestion in remote rural areas.

VIII. CONCLUSION & OPEN SCIENCE ARTIFACTS

This paper presented **BharatGov Access**, a census-scale web observatory evaluating 12,092 Indian government portals through an in-country domestic vantage point in Bangalore. By establishing a standardized 5-Pillar / 10-Sub-Feature framework, we have demonstrated the national accessibility dividends of NIC's S3WaaS platform while revealing critical security gaps in HTTP defense headers.

To promote reproducibility and open science, all research artifacts are made publicly accessible:

- **Interactive Web Observatory:** `https://quality-ampland-ashley-ought.trycloudflare.com`
- **Census Open Dataset:** Immutable Apache Parquet, CSV, and JSON releases hosted under CC-BY-4.0 in `resources/bharatgov_access_clean.*`.

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